The KARIS - Project

K-itting
A-utomation
R-obots
(in)
I-ndustrial
S-ettings

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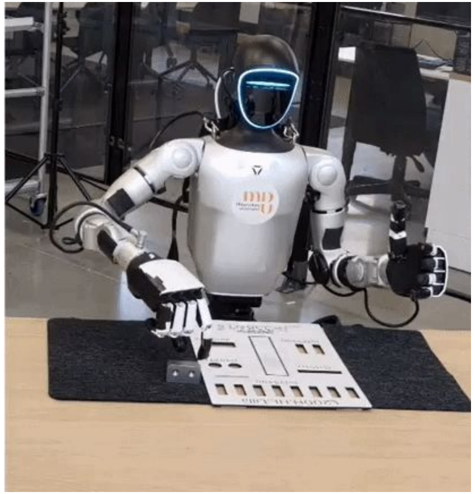
Why humanoid robots?

- Industrial environments are primarily designed for humans.
- Traditional automation often requires dedicated equipment or workspace modifications.
- Humanoid robots could potentially use existing workstations and tools.

Building on previous iterations

- Previous attempts have used telescoping.
- VLA control for both movement and detection.
- Object detection and manipulation.
- Previous solution was slow and unreliable.



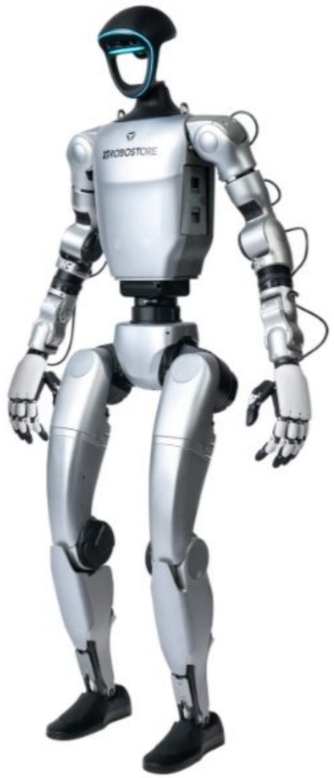
Project goal - Demonstrate autonomous execution of a kitting task.

The robot should:

1. Detect known objects
2. Classify the objects
3. Estimate their position and orientation
4. Pick them up
5. Move them to the correct location
6. Place them on the kitting tray

G1 (U6) EDU Ultimate D

- Stereo-vision camera in combination with a 3D LiDAR.
- FTP dexterous hands for precision tasks.
- Jetson Orin.
- 3kg arm max load.

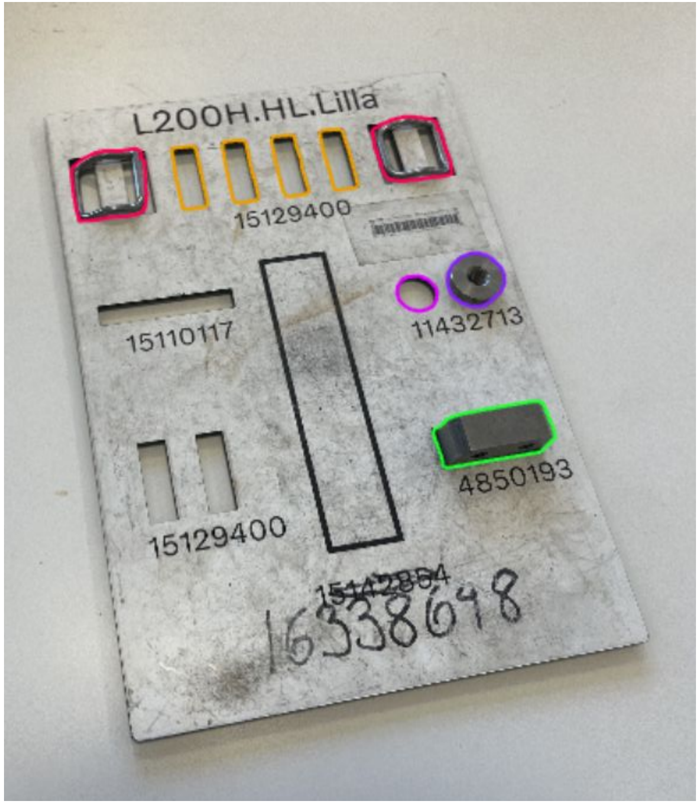


Initial approach and ideas

- YOLO - Object detection.
- FoundationPose - Pose estimation.
- MoveIt2 - Motion planning.
- Pinocchio - Balance.
- IsaacLab

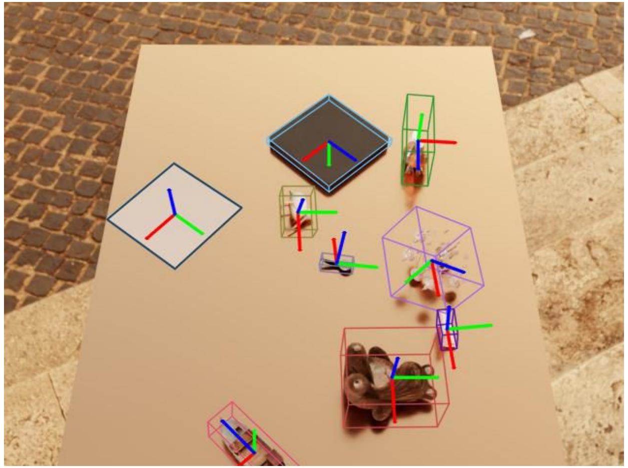
YOLO - based object detection

- Real-time object detection algorithm.
- Identifies what objects are present and where they are.
- Fast and suitable for real-time robotic applications.



FoundationPose - Pose estimation

- Uses RGB and depth data to understand the scene.
- Matches the observed object to a 3D model.
- Enables accurate grasping.



MoveIt2

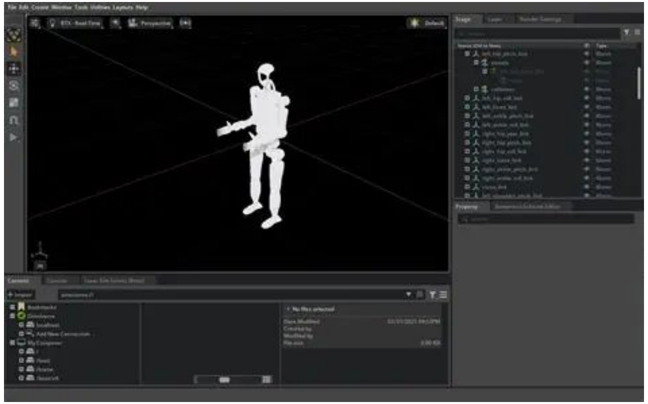
- Calculates a path from the robot’s current position to the target.
- Avoids collisions.
- Ensures the robot does not hit itself or surrounding objects.
- Converts the planned path into movements.

Pinocchio

- Creates a representation of joints, links, and their connections.
- Essential in rigid body dynamics and inverse dynamic calculations.
- Determines the position and orientation of the robot’s joints.

IsaacLab

- Simulation software for robot learning
- Realistic physics
- Collision handling
- Sensor simulation



Project plan and division of work

The project is divided into 3 separate but integral parts:

- Perception
  - Object detection
  - Object classification
  - Pose estimation
  - Sensor processing
- Manipulation
  - Inverse kinematics
  - Motion planning
  - Grasping and placing.
- Integration
  - ROS Architecture
  - Coordinate transformations
  - Communication

Expected outcome

Detect — Pick up — Move — Place

In a lab environment

Objects clearly separated. Known workspace in controlled conditions

Manipulate the chosen object

Gripping the object. Using predetermined grip-points.

Moving, rotating, fitting

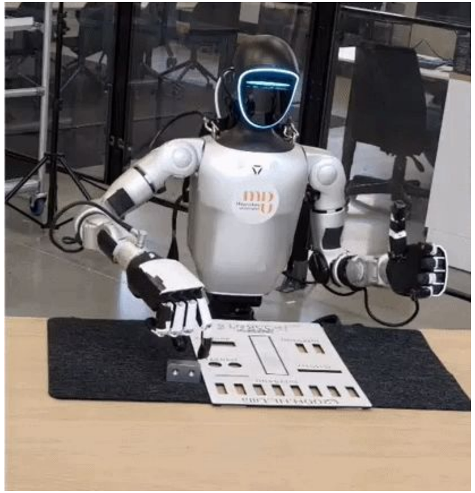
Picking up the object. Moving towards the slotted location.

Letting go safely

Verify the task as complete. Making adjustments.

Project demo plan

- Movement of the unitree G1 in a controlled environment.
- Detection of certain objects.
- Completing a kitting task in an ideal scenario.



Future

- Build on previous master’s thesis work
- Divide the system into specialized modules
- Move away from an all-encompassing VLA approach
- Long-term goal: robots that can perform complex tasks with human-like ease